

HCI Group Project

Title: Heuristic Evaluation and Possible Redesign of an Interactive System

Deadline: June 10, 2026

Poster Presentation: June 11, 2026

Groups: You will work on the project in groups of 3-5 people

Objective: In this project, you will conduct heuristic evaluation of the usability of an interactive system against Jakob Nielsen's heuristics for Interaction design. To check compliance with a specific rule, ask relevant questions:

1. Visibility of System Status:

- Does the design clearly communicate its state?
- Is feedback presented quickly after user actions?

2. Match Between System and Real World:

- Will user be familiar with the terminology used in the design?
- Do the design's controls follow real-world conventions?

3. User Control and Freedom:

- Does the design allow users to go back a step in the process?
- Are exit links easily discoverable?
- Can users easily cancel an action?
- Is Undo and Redo supported?

4. Consistency and Standards:

- Does the design follow industry conventions?
- Are visual elements used consistently throughout the design?
(Do similar system elements look similar?)

5. Error Prevention: Minimize the likelihood of making mistakes.

- Does the design prevent slips by using helpful constraints?
- Does the design warn users before they perform risky actions?

6. Recognition Rather Than Recall:

- Does the design keep important information visible, so that users do not have to memorize it?
- Does the design offer help in-context?

7. Flexibility and Efficiency of Use: Both new and experienced users should be able to efficiently use the system.

- Does the design provide accelerators like keyboard shortcuts and touch gestures?
- Is content and functionality personalized or customized for individual users?

8. Aesthetic and Minimalist Design: Declutter. Less is more.

- Is the visual design and content focused on the essentials?
- Have all distracting, unnecessary elements been removed?

9. Help Users Recognize, Diagnose, and Recover from Errors:

- Does the design use traditional error message visuals, like bold, red text?
- Does the design offer a solution that solves the error immediately?

10. Help and Documentation:

- Is help documentation easy to search?
- Is help provided in context right at the moment when the user requires it?

You will identify usability issues in the system's interface based on these heuristics and propose a redesign (if the system violates one or more of these heuristics) to enhance usability and user experience. You will present

your analysis and proposed interface redesign through a poster presentation.

Note: you will propose a redesign of the user interface only if your evaluation of the system reveals issues in the user interface according to the selected heuristics.

Project Stages:

1. System Selection:

- You will select an existing interactive system to analyze and possibly redesign. This system could be a website, mobile application, software application, an appliance, a kiosk, a wearable device, or any other interactive platform.

2. Heuristic Evaluation of the system interface:

- You will identify usability issues within the current user interface based on Nielsen's usability principles. For example, you might observe inconsistencies in interface elements, confusing navigation, or lack of feedback for user actions.
- First, spend some time using the interface to understand its primary functions and overall structure.
- Second, evaluate the interface against Nielsen's heuristics. Look for specific places where the interface fails to adhere to the guideline.
- Assign severity rating to each usability issue discovered during the evaluation (e.g., cosmetic, minor, major, catastrophe)

3. Heuristic Evaluation of a task:

- Choose a specific, representative task that a user would perform on the interface.
- Understand task flow on the interface. Understand the individual steps of the task.
- Evaluate each step of the task against Nielsen's heuristics. At each step, look for specific places where the interface fails to adhere to the heuristics.

- Assign severity rating to each usability issue (e.g., cosmetic, minor, major, catastrophe)

4. Interface Redesign:

- Based on your analysis, you will propose a redesign of the system's interface to address the identified usability issues and to improve user experience.
- The redesign proposal should describe how it resolves the usability issues identified during the evaluation.

5. Poster Presentation:

- You will create a poster to summarize the findings of your evaluation of the current system and to present the proposed user interface redesign.
- The poster should include visual representations of the current interface (screenshots or sketches), highlighting areas of concern, as well as sketch(es) of the proposed redesign.
- You will present your posters in the class, explaining their analysis findings and rationale behind the proposed redesign.

Deliverables:

The deliverable is the poster itself which should include the following sections:

- Title
- Group Members (IDs and pinyin Names)
- Description of the selected system (purpose and key functions/features)
- Objectives of the Evaluation
- Evaluation Methodology
- Results (including images, screenshots, explanations)
- A prioritized list of usability issues.
- Proposed UI Improvements/Redesign (if applicable)